

Beyond Searching a Village: Learning to Recommend Diverse and Successful Collaborative Teams

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Abstract

Team recommendation involves selecting skilled experts to form an *almost surely* successful collaborative team, or refining the team composition to maintain or excel at performance. To address the tedious and error-prone manual process, computational approaches have been proposed, especially for web-scale social networks and widespread online collaboration and diversity of interactions. In this tutorial, with a brief overview of pioneering subgraph optimization approaches and their shortfalls, we deliberately focus on the recent learning-based approaches, with a particular in-depth exploration of graph neural network-based methods. More importantly, we then discuss team *refinement*, which involves structural adjustments or expert replacements to enhance team performance in dynamic environments. Finally, we discuss training strategies, benchmarking datasets, and open-source libraries, along with future research directions and real-world applications. Further resources are available at <https://fani-lab.github.io/OpENTF/tutorial/recsys26>.

CCS Concepts

• General and reference → Surveys and overviews; • Information systems → Recommender systems; Social recommendation; • Human-centered computing → Social recommendation; • Computing methodologies → Neural networks; • Theory of computation → Graph algorithms analysis.

Keywords

Team Recommendation; Social Information Retrieval; OpENTF; Adila

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1 Introduction

From Figure 1, team recommendation aims to automate forming collaborative teams of experts capable of successfully accomplishing a given task, and its applications span a wide range of domains. In emergency response, it involves assembling ad hoc teams of diverse

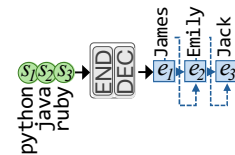


Figure 1: Team recommendation.

[10 min]	Pioneering Techniques
[10 min]	Subgraph Optimization Objectives and Techniques
[60 min]	Learning Team Recommendation
[35 min]	Team Refinement
[15 min]	Structural Adjustment
[20 min]	Membership Substitution
[20 min]	Training Strategies
[10 min]	Evaluation Methodologies
[05 min]	Datasets
[05 min]	Effectiveness/Efficiency
[15 min]	Future Directions
[05 min]	Fair and Diverse Team Recommendation
[05 min]	Spatial Team Recommendation
[05 min]	Multi-Objective Neural Team Recommendation
[20 min]	Real-World Applications
[10 min]	OpENTF and Adila

Figure 2: Our tutorial's timeline.

professionals while accounting for expertise, communication efficiency, availability, and logistical constraints. In scientific publishing, it supports the assignment of peer reviewers, where identifying an appropriate review panel remains a bottleneck due to the manual selection process. In education, educators seek to partition students into collaborative teams that promote peer learning, engagement, and social skill development, a need that has become increasingly important with the growth of large-scale online classes. On online freelance platforms, where abundant information is available about experts' skills, past performance, and collaboration history, team recommendation enables forming effective teams for numerous projects with evolving requirements.

Historically, teams were formed manually based on human experience and instinct; a tedious, error-prone, and suboptimal process due to unfair biases, multiple criteria to optimize like budget, time and team size, and a large number of candidates, which makes it all the more difficult. Hence, researchers have long been developing computational models that learn relationships between experts and their skills to recommend *almost surely* successful teams, and/or refine existing teams to maintain or even improve their success.

2 Prior Tutorials

Despite the abundance, there had been *no* comprehensive tutorial on team recommendation methods until recently when we began to fill the gap by tutorials at UMAP24¹ [21], centered on *online*

¹<https://www.um.org/umap2024/tutorials/>



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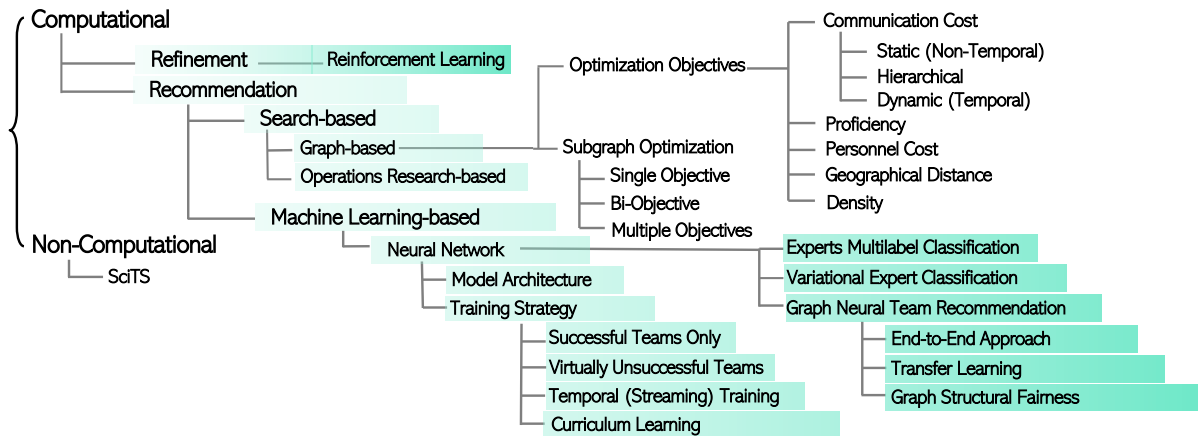


Figure 3: Taxonomy of team recommendation methods.

skilled users, SIGIR-AP24² [22], based on a novel taxonomy from a computational perspective with a focus on real-world applications, and at WSDM25³ [23] and CIKM25⁴ [19], to review seminal solutions with a special focus on the emerging graph neural network-based methods. In this tutorial, from Figure 2 and 3, we expand on them based on our recently published [20] and ongoing surveys of neural approaches that leverage graph neural networks [16, 27], sequence-to-sequence models and transformers [26]. Notably, we further investigate team refinement methods, such as expert replacement through reinforcement learning in dynamic sport teams or online video games [13, 28]. We also present methods to mitigate inherent biases via curriculum learning and loss regularization [4, 11, 15].

3 Targeted Audience

We target *beginner* or *intermediate* researchers, industrial practitioners with a broad interest in developing web-scale recommender systems, willing to have a whole picture of team recommendation techniques. Furthermore, we target the graph neural network community for a comprehensive review of subgraph optimization objectives and call them for further development of effective yet efficient graph machine learning frameworks with a special focus on team recommendation. Last, this tutorial enables organizations and practitioners to compare different approaches and readily pick the suitable one for their application to form or maintain successful teams. The target audience will be those *familiar* with graph theory and neural architectures. We provide sufficient details about advanced techniques so that the content becomes accessible to those with a fair knowledge of fundamentals.

4 Tutorial Summary (180 minutes)

Based on a taxonomy of computational methods, as shown in Figure 3, we start our tutorial with a brief introduction to the pioneering graph-based team recommendation algorithms [9], then continue to explore the learning-based team recommendation [1, 4, 7, 8, 10, 16–18, 24, 26] and team refinement methods [3, 13, 28, 29], focusing on modern graph neural networks and reinforcement learning. We explain strategies to train team recommenders, including negative sampling [6], temporal training [8], and curriculum

learning [4]. We also discuss the benchmark datasets, what has been considered successful for a team as the ground truth, and metrics utilized to measure the quality of the recommended teams.

Toward the end, we discuss open issues and future directions, including:

- **Fair and Diverse Teams:** Existing methods heavily influenced by hidden *unfair* societal biases [11, 12, 15]. Meanwhile, social science research provides compelling evidence about the synergistic effects of diversity on team performance. In our tutorial, we introduce notions of fairness and protected attributes and study debiasing algorithms to mitigate the potential unfairness in the models' recommendations.
- **Spatial Team Recommendation:** In search of an optimal team, companies further look for experts in a region where the organization is *geographically* based. We bring forth the *spatial* team recommendation problem; that is, given a set of experts, skills and geolocations, the goal is to determine if the combination of skills and geolocations in forming teams has synergistic effects.
- **Multi-Objective Optimization:** In real-world scenarios, balancing multiple, often conflicting objectives (e.g., team effectiveness and experts' workload distribution) requires a training process guided by a loss function that explicitly accounts for multiple objectives. However, existing approaches commonly frame the problem as a classification or link prediction task, aim to maximize the coverage of the required skills, and overlook designing a task-aware loss function.

Moreover, we explain novel applications of team recommendation, including learning group recommendation [5, 14], peer-review assignments [2], and palliative care [25].

Finally, we present OpeNTF⁵ [7, 10], a benchmark library for neural models that can efficiently preprocess large-scale datasets, can be easily extended or customized to new models, and is extensible to new datasets from other domains. We also introduce Adila⁶ [11], which enables further post-processing reranking to the list of recommended experts to ensure a fair outcome, and vivafemme [15], an in-processing method for female-advocate loss regularization during model training.

²<https://www.sigir-ap.org/sigir-ap-2024/tutorial/>

³<https://www.wsdm-conference.org/2025/tutorials/>

⁴<https://cikm2025.org/program/tutorials>

⁵<https://github.com/fani-lab/OpeNTF>

⁶<https://github.com/fani-lab/Adila>

References

- [1] Md Jamil Ahmed, Mahdis Saeedi, and Hossein Fani. 2024. Vector Representation Learning of Skills for Collaborative Team Recommendation: A Comparative Study. In *Web Information Systems Engineering - WISE 2024 - 25th International Conference, Doha, Qatar, December 2-5, 2024, Proceedings, Part II (Lecture Notes in Computer Science, Vol. 15437)*, Mahmoud Barhamgi, Hua Wang, and Xin Wang (Eds.). Springer, 193–207. doi:10.1007/978-981-96-0567-5_15
- [2] Meltem Aksoy, Seda Yanik, and Mehmet Fatih Amasyali. 2023. Reviewer Assignment Problem: A Systematic Review of the Literature. *J. Artif. Intell. Res.* 76 (2023), 761–827. doi:10.1613/JAIR.1.14318
- [3] Mohammed Almutairi, Charles Chiang, Yuxin Bai, and Diego Gómez-Zarà. 2025. tAlfa: Enhancing Team Effectiveness and Cohesion with AI-Generated Automated Feedback. In *Proceedings of the 4th Annual Symposium on Human-Computer Interaction for Work, CHIWORK 2025, Amsterdam, Netherlands, June 23-25, 2025*. ACM, 2:1–2:25. doi:10.1145/3729176.3729197
- [4] Reza Barzegar, Marco Nikola Kurepa, and Hossein Fani. 2025. Adaptive Loss-based Curricula for Neural Team Recommendation. In *Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining, WSDM 2025, Hannover, Germany, March 10-14, 2025*, Wolfgang Nejdl, Sören Auer, Meeyoung Cha, Marie-Francine Moens, and Marc Najork (Eds.). ACM, 914–923. doi:10.1145/3701551.3703574
- [5] Gonzalo Candel, Víctor Sánchez-Anguix, Juan M. Alberola, Vicente Julián, and Vincent J. Botti. 2023. An Integer Linear Programming Model for Team Formation in the Classroom with Constraints. In *Hybrid Artificial Intelligent Systems - 18th International Conference, HAIS 2023, Salamanca, Spain, September 5-7, 2023, Proceedings (Lecture Notes in Computer Science, Vol. 14001)*, Pablo García Bringas, Hilde Pérez García, Francisco Javier Martínez de Pisón, Francisco Martínez-Álvarez, Alicia Troncoso Lora, Álvaro Herrero, José Luis Calvo-Rolle, Héctor Quintián, and Emilio Corchado (Eds.). Springer, 397–408. doi:10.1007/978-3-031-40725-3_34
- [6] Arman Dashti, Saeed Samet, and Hossein Fani. 2022. Effective Neural Team Formation via Negative Samples. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management, Atlanta, GA, USA, October 17-21, 2022*, Mohammad Al Hasan and Li Xiong (Eds.). ACM, 3908–3912. doi:10.1145/3511808.3557590
- [7] Arman Dashti, Karan Saxena, Dhvani Patel, and Hossein Fani. 2022. OpeNTE: A Benchmark Library for Neural Team Formation. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management, Atlanta, GA, USA, October 17-21, 2022*, Mohammad Al Hasan and Li Xiong (Eds.). ACM, 3913–3917. doi:10.1145/3511808.3557526
- [8] Hossein Fani, Reza Barzegar, Arman Dashti, and Mahdis Saeedi. 2024. A Streaming Approach to Neural Team Formation Training. In *Advances in Information Retrieval - 46th European Conference on Information Retrieval, ECIR 2024, Glasgow, UK, March 24-28, 2024, Proceedings, Part I (Lecture Notes in Computer Science, Vol. 14608)*, Nazli Goharian, Nicola Tonello, Yulan He, Aldo Lipani, Graham McDonald, Craig Macdonald, and Iadh Ounis (Eds.). Springer, 325–340. doi:10.1007/978-3-031-56027-9_20
- [9] Theodoros Lappas, Kun Liu, and Evimaria Terzi. 2009. Finding a team of experts in social networks. In *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Paris, France, June 28 - July 1, 2009*, John F. Elder IV, Françoise Fogelman-Soulié, Peter A. Flach, and Mohammed Javeed Zaki (Eds.). ACM, 467–476. doi:10.1145/1557019.1557074
- [10] Hamed Loghmani, Md Jamil Ahmed, Kap Thang, Gabriel Rueda, and Hossein Fani. 2026. OpeNTE2: Fairness-aware Graph Neural Team Formation. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2026, Melbourne/VIC Australia, July 20-24, 2026*. ACM, 3300–3309. doi:10.1145/3805712.3808605
- [11] Hamed Loghmani and Hossein Fani. 2023. Bootless Application of Greedy Re-ranking Algorithms in Fair Neural Team Formation. In *Advances in Bias and Fairness in Information Retrieval - 4th International Workshop, BIAS 2023, Dublin, Ireland, April 2, 2023, Revised Selected Papers (Communications in Computer and Information Science, Vol. 1840)*, Ludovico Boratto, Stefano Faralli, Mirko Marras, and Giovanni Stilo (Eds.). Springer, 108–118. doi:10.1007/978-3-031-37249-0_9
- [12] Hamed Loghmani, Mahdis Saeedi, Gabriel Rueda, Edwin Paul, and Hossein Fani. 2026. A Probabilistic Greedy Attempt to Be Fair in Neural Team Recommendation. *Computational Intelligence* 42, 3 (2026), e70221. doi:10.1111/coin.70221
- [13] Bing Lv, Junji Jiang, Likang Wu, and Hongke Zhao. 2024. Team formation in large organizations: A deep reinforcement learning approach. *Decis. Support Syst.* 187 (2024), 114343. doi:10.1016/J.DSS.2024.114343
- [14] Larry K Michaelsen, Arletta Bauman Knight, and L Dee Fink. 2023. *Team-based learning: A transformative use of small groups in college teaching*. Taylor & Francis.
- [15] Roonak Moasses, Delaram Rajaei, Hamed Loghmani, Mahdis Saeedi, and Hossein Fani. 2024. vivafemme: Mitigating Gender Bias in Neural Team Recommendation via Female-Advocate Loss Regularization. In *Advances in Bias and Fairness in Information Retrieval - 5th International Workshop, BIAS 2024, Washington, DC, USA, July 18, 2024, Vol. 2227*. Springer, 78–90. doi:10.1007/978-3-031-71975-2_6
- [16] Radin Hamidi Rad, Hossein Fani, Ebrahim Bagheri, Mehdi Kargar, Divesh Srivastava, and Jaroslav Szlichta. 2024. A Variational Neural Architecture for Skill-based Team Formation. *ACM Trans. Inf. Syst.* 42, 1 (2024), 7:1–7:28. doi:10.1145/3589762
- [17] Radin Hamidi Rad, Hossein Fani, Mehdi Kargar, Jaroslav Szlichta, and Ebrahim Bagheri. 2020. Learning to Form Skill-based Teams of Experts. In *CIKM '20: The 29th ACM International Conference on Information and Knowledge Management, Virtual Event, Ireland, October 19-23, 2020*. ACM, 2049–2052. doi:10.1145/3340531.3412140
- [18] Radin Hamidi Rad, Shirin Seyedalehi, Mehdi Kargar, Morteza Zihayat, and Ebrahim Bagheri. 2022. A Neural Approach to Forming Coherent Teams in Collaboration Networks. In *Proceedings of the 25th International Conference on Extending Database Technology, EDBT 2022, Edinburgh, UK, March 29 - April 1, 2022*, Julia Stoyanovich, Jens Teubner, Paolo Guagliardo, Milos Nikolic, Andreas Pieris, Jan Mühlhlig, Fatma Özcan, Sebastian Schelter, H. V. Jagadish, and Meihui Zhang (Eds.). OpenProceedings.org, 2:440–2:444. doi:10.48786/EDBT.2022.37
- [19] Mahdis Saeedi and Hossein Fani. 2025. Neural Shifts in Collaborative Team Recommendation. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management, CIKM 2025, Seoul, Republic of Korea, November 10-14, 2025*, Meeyoung Cha, Chanyoung Cha, Noseong Park, Carl Yang, Senjuti Basu Roy, Jessie Li, Jaap Kamps, Kijung Shin, Bryan Hooi, and Lifang He (Eds.). ACM, 6845–6848. doi:10.1145/3746252.3761456
- [20] Mahdis Saeedi, Hawre Hosseini, Christine Wong, and Hossein Fani. 2025. A Survey of Subgraph Optimization for Expert Team Formation. *ACM Comput. Surv.* 57, 12 (2025), 314:1–314:40. doi:10.1145/3737455
- [21] Mahdis Saeedi, Christine Wong, and Hossein Fani. 2024. Collaborative Team Recommendation for Skilled Users: Objectives, Techniques, and New Perspectives. In *Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization, UMAP Adjunct 2024, Cagliari, Italy, July 1-4, 2024*. ACM. doi:10.1145/3631700.3658521
- [22] Mahdis Saeedi, Christine Wong, and Hossein Fani. 2024. Paradigm Shifts in Team Recommendation: From Historical Subgraph Optimization to Emerging Graph Neural Network. In *Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region, SIGIR-AP 2024, Tokyo, Japan, December 9-12, 2024*, Tetsuya Sakai, Emi Ishita, Hiroaki Ohshima, Faegheh Hasibi, Jiaxin Mao, and Joemon M. Jose (Eds.). ACM, 295–298. doi:10.1145/3673791.3698440
- [23] Mahdis Saeedi, Christine Wong, and Hossein Fani. 2025. Bridging Historical Subgraph Optimization and Modern Graph Neural Network Approaches in Team Recommendation. In *Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining, WSDM 2025, Hannover, Germany, March 10-14, 2025*, Wolfgang Nejdl, Sören Auer, Meeyoung Cha, Marie-Francine Moens, and Marc Najork (Eds.). ACM, 1016–1019. doi:10.1145/3701551.3703483
- [24] Mohammad Hossein Saliminabi, Sajad Ebrahimi, Radin Hamidi Rad, Dimitrios Androustos, Fattane Zarrinkalam, and Ebrahim Bagheri. 2026. Diffusion-based generative modeling for expert team formation. *Inf. Process. Manag.* 63, 7 (2026), 104862. doi:10.1016/J.IJPM.2026.104862
- [25] Kalyani Selvarajah, Pooya Moradian Zadeh, Ziad Kobti, Mehdi Kargar, Mohd Tazim Ishraque, and Kathryn Pfaff. 2018. Team Formation in Community-Based Palliative Care. In *2018 Innovations in Intelligent Systems and Applications, INISTA 2018, Thessaloniki, Greece, July 3-5, 2018*. IEEE, 1–7. doi:10.1109/INISTA.2018.8466272
- [26] Kap Thang, Hawre Hosseini, and Hossein Fani. 2025. Translative Neural Team Recommendation: From Multilabel Classification to Sequence Prediction. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2025, Padua, Italy, July 13-18, 2025*, Nicola Ferro, Maria Maistro, Gabriella Pasi, Omar Alonso, Andrew Trotman, and Suzan Verberne (Eds.). ACM, 2800–2806. doi:10.1145/3726302.3730259
- [27] Karan Vombatkere, Theodoros Lappas, and Evimaria Terzi. 2025. A QUBO Framework for Team Formation. In *Machine Learning and Knowledge Discovery in Databases. Research Track - European Conference, ECML PKDD 2025, Porto, Portugal, September 15-19, 2025, Proceedings, Part II (Lecture Notes in Computer Science, Vol. 16014)*, Rita P. Ribeiro, Bernhard Pfahringer, Nathalie Japkowicz, Pedro Larrañaga, Alipio M. Jorge, Carlos Soares, Pedro H. Abreu, and João Gama (Eds.). Springer, 492–510. doi:10.1007/978-3-032-05981-9_29
- [28] Yifan Zang, Jinmin He, Kai Li, Haobo Fu, Qiang Fu, Junliang Xing, and Jian Cheng. 2023. Automatic Grouping for Efficient Cooperative Multi-Agent Reinforcement Learning. In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine (Eds.). http://papers.nips.cc/paper_files/paper/2023/hash/906c860f1b7515a8ffec02dcac74048-Abstract-Conference.html
- [29] Sharon Zhou, Melissa A. Valentine, and Michael S. Bernstein. 2018. In Search of the Dream Team: Temporally Constrained Multi-Armed Bandits for Identifying Effective Team Structures. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, CHI 2018, Montreal, QC, Canada, April 21-26, 2018*, Regan L. Mandryk, Mark Hancock, Mark Perry, and Anna L. Cox (Eds.). ACM, 108. doi:10.1145/3173574.3173682